

Project Design Phase-II

Technology Stack

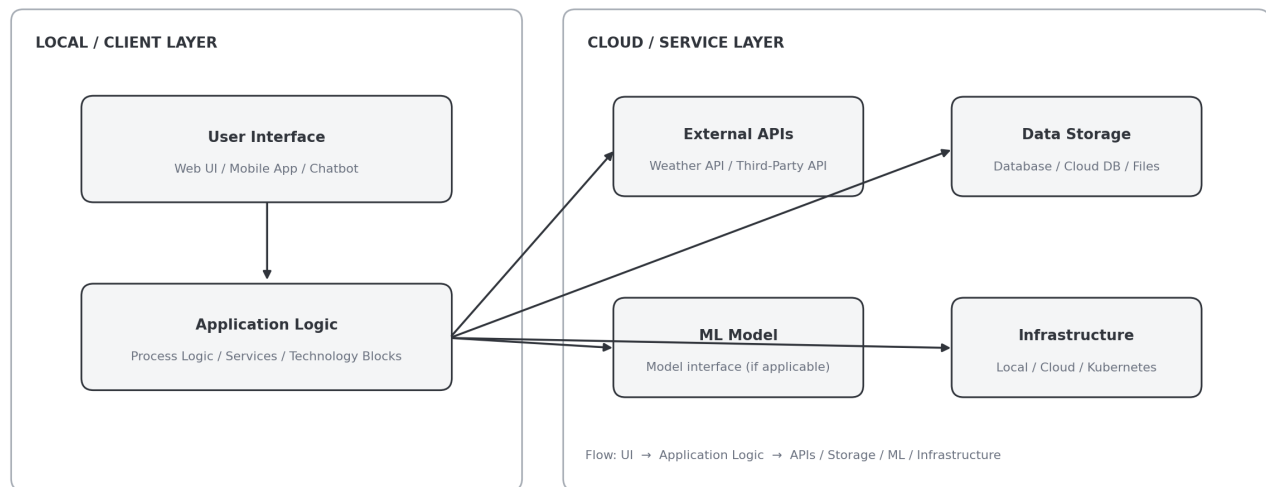
(Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-7291
Project Name	Weather Wise
Maximum Marks	4 Marks

1. Technical Architecture

Present the solution architecture as a clear diagram, together with the supporting information listed in Table 1 and Table 2. The diagram should show how the application components connect and where each technology is positioned.

AI WEATHER WISE - TECHNICAL ARCHITECTURE



Guidelines

- Show every process as an application-logic or technology block.
- Clearly separate local infrastructure from cloud infrastructure.
- Display all external connections, including third-party APIs.
- Include the relevant data-storage components or services.
- Show the interface to machine-learning models wherever applicable.

2. Table 1 - Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Defines how users interact with the application, for example through a Web UI, mobile app, or chatbot.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2	Application Logic-1	Represents the logic used to execute a process within the application.	Java / Python
3	Application Logic-2	Represents the logic or service used for another process in the application.	IBM Watson STT service
4	Application Logic-3	Represents the logic or service used for another process in the application.	IBM Watson Assistant
5	Database	Specifies the required data types, database setup, and configuration details.	MySQL, NoSQL, etc.
6	Cloud Database	Identifies the database service hosted in the cloud environment.	IBM DB2, IBM Cloudant etc.
7	File Storage	Defines the storage needed for files used by the application.	IBM Block Storage or Other Storage Service or Local Filesystem
8	External API-1	States the purpose of the first external API integrated with the application.	IBM Weather API, etc.
9	External API-2	States the purpose of the second external API integrated with the application.	Aadhar API, etc.
10	Machine Learning Model	Explains the role of the machine-learning model used by the solution.	Object Recognition Model, etc.
11	Infrastructure (Server / Cloud)	Describes whether the application is deployed locally or in the cloud, including local and cloud server configuration.	Local, Cloud Foundry, Kubernetes, etc.

3. Table 2 - Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks included in the application.	Technology of Opensource framework
2	Security Implementations	List the implemented security and access-control measures, including firewall usage where relevant.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3	Scalable Architecture	Explain how the architecture supports scalability, such as through a 3-tier design or micro-services.	Technology used
4	Availability	Explain how application availability is maintained, for example using load balancers or distributed servers.	Technology used
5	Performance	Describe the performance design considerations, such as requests per second, caching, and CDN usage.	Technology used

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